

Preprocessing and Analysing Daily Sales: Exploring the Predictive Potential of LASSO and
RIDGE Regression.

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Abstract

In this report, we handled a large dataset of about 1 million observations describing the retail sales of 1115 drug stores across Germany in the years 2013 until 2015. The primary objective is to predict the sales of the stores given in a test-set file starting from August of 2015 and ending on the 17th of September of 2015. During this process, useful analyses about the importance of variables occurred and we revealed several insights about the impact of promotions and competition on the sales. The preprocessing of the data exploited several statistical and data-science methods to manage data, treating missing values, generating and extracting new features. We used summarizing technics to reveal trends and insights into the data and performed correlation and regression analyses to estimate the importance of the variables. We used simple and advanced multivariate linear regression models to analyse the data, perform feature selection and model optimization using also cross-validation (CV) technics. We use the Python programming language with (a) the *sklearn* module which includes state-of-the-art models of RIDGE and LASSO regression models presenting also automated CV implementations and (b) the implementation of the R's *statsmodels* API. We used a validation set of one month to evaluate our predictive model which achieved an R2 score of 0.74 versus an R2 of 0.84 on the training set. Finally, we predicted the sales for the required period, perform adjustments on the predictions and saved them in a CSV file for further evaluation. We report all the above results with plots and tables.

Keywords: Linear Regression, Lasso Regression, Ridge Regression, Feature Selection, Cross-Validation, Predictive model, data wrangling, data preprocessing, Sales Forecasting.

Preprocessing and Analysing Daily Sales: Exploring the Predictive Potential of LASSO and RIDGE Regression.

Forecasting retail sales is a demanding task of the data science and statistics fields. Sales height can contain a severe amount of noise and randomness may play a decisive role in the outcome. Forecasting sales could be crucial for managers to estimate the number of employees, the amount and quality of products, the schedule of orders, and a logistics plan. Sales are sensitive to seasonality but also can be sensitive to the quality and availability of products. Managers need to estimate sales to perform financial programming of their business so an accurate predictive model is of high importance to companies related to trading.

Data is the most important thing when dealing with forecasting models. Data not only should be accurate but also should be adequate for the proposed task. Even a small portion of impaired data may poison the forecasting model but even a few rows of missing data may be the reason for not achieving the desired result. Data scientists should balance the missing data and the probability of it being poisonous for the model. On the one hand, data scientists should try to keep as many data as possible and on the other hand, they should be careful to discover and handle outliers and poisonous observations. Handling data though frequently underestimated, can be a challenging task demanding domain knowledge and statistical literacy.

A predictive model needs explanatory variables. Nevertheless, dimensionality curse is a known drawback to predictive models so several technics can be used to drop variables from the models. Before starting analysing data and training a model, a bunch of variables can be presented as being relative to the target but this is not always the case. A model should not use every possible variable that may look relative. Some variables can be also misleading, unrelated to the target, or even present a random correlation in the examined set but not existed in real life. The task of selecting variables is known as Feature Selection and methods to analyze the importance of variables and come to a decision to keep it or drop it may include (a) univariate regression analysis (b) p-value (c) testing with a model. In this study, we use LASSO regression and p-values of variables to perform a feature selection.

This study is limited to linear regression models. The state-of-the-art linear regression models penalize the coefficients performing that way a regularization. The LASSO regression can penalize coefficients to the point of zero, performing that way a feature selection and RIDGE shrink coefficients of the features but without zeroing them. We explore both of these regressors in different tasks and different data structures. We use the LASSO variant of Larson to perform feature selection in a dataset not taking direct account of stores and then we train our model with RIDGE regression using a dataset containing one-hot-encoded stores. The python sklearn module wraps cross-validation in their RIDGE and LASSO algorithms thus automating the process of model optimization.

Our model selection method includes the training of the RIDGE model and a multivariate linear model with the exact predictors but without regularization. We select the most efficient model after comparing a series of metrics like Mean Absolute Error (MAE), Mean Absolute

Percentage Error (MAPE), Root Mean Squared Error (RMSE), Root Mean Squared Percentage Error (RMSPE) and the Correlation Coefficient (R^2).

Methodology

In this study, we propose a pipeline of processes and actions (Figure 1) that could maximize the efficiency of preprocessing and training the model. Starting from loading data from the database we suggest a high-level plan for the project. A predefined order of actions after loading the data can minimize errors and delinquencies. Specifically, after loading the data we propose to calculate the basic statistics and seek for outliers. Then, we suggest cleaning the data from inconsistencies and treat the missing values. A feature generation, feature extraction and data wrangling process may follow and after completing these steps a series of analyses could be conducted. A regression task like this one can find use with correlation analysis, followed by regression analysis and ending with a features selection. A model selection process can follow these steps and then project ends with the prediction section. A more complicated implementation of the above plan can include going back and forth in the pipeline as long as the result keeps improving.

Data¹

We use the panda's module of the Python programming language to load and handle the data and perform basic statistics. Our data came in a Comma Separated Value (CSV) structure consisting of 3 datasets. The first one comprises our training data and consists of 1,017,209 observations regarding daily sales of 1115 drug-stores from 2013 until the end of July of 2015. The training set contains 9 variables like an id of the store, the day of the week as a series of numbers (1-7) where 7 equals Sunday and 1 equals Monday, the sales of a store on a date, the number of customers at this date, and four binary variables to indicate whether the store is open, the store is under promotion, the date is a state holiday, and the date is a school holiday. The target variable of the project is the sales variable and the remaining variables can be the predictors. The second dataset comprises the test set and contains 41,088 observations. The dataset misses the information about sales which is a variable to be predicted and the number of customers. The sales to be predicted occur in the period starting from the 1st of August 2015 and end on the 17th of September of 2015. The third dataset contains extra information for the 1115 stores of the training data. These explanatory variables are related to the nature of each store, the competition distances and history, and another extra promotion that may run on certain periods. This promotions in the third set will be identified from now on as Promo2 while the promotions in the training set will be identified as Promo.

Basic Statistics.

The basic statistics (Table 1) show that the train set and the test set share similar values of the mean and the standard deviation of three variables. These are the DayOfWeek, the Promo and the Open variables while there is a significant difference in the SchoolHoliday variable where the mean of the train set is 0.18 and the mean of the test set is 0.44 revealing that in the test period

the school holidays are almost the half days. Moreover, though the store id (Store) is a categorical variable we expect the mean of stores id to be identical indicating that the ids are distributed analogically in the observations, something which did not occur. This means that the presence of stores in test and train set could differ or even the stores are not the same ones in both sets.

Outliers.

We use basic statistics to detect some obvious outliers in the data. In this case, there was a strange value in the stores' dataset in the competition variable describing the age of the competition. A minimum value of the year 1900 was presented while 75% of the competition starts after the year 2006 and the mean year of the variable is 2008. A further investigation of the years shows that there are two years (1900, 1961) very far from the next closest year of the distribution (1990). Specifically, the 10 lowest observations of this variable are: [1900, 1961, 1990, 1994, 1995, 1998, 1999, 2000, 2001, 2002]. We cannot be certain that there is not a competition bornt 120 years ago, but such distant observations might skew the result in a regression task so we decided to treat these two values as outliers.

We deducted 4 standard deviations of the distribution (std: 6.19) from the mean (2008) to calculate the minimum accepted year ($2008 - 4 * 6.19 = \text{int}(1983)$) that would not skew the regression. So, we replaced the year 1900 with 1983. Then we added one standard deviation to the calculated minimum value ($1983 + 6.19$) and replaced the year 1961 with the floor of this sum (1989).

Cleaning.

Data cleaning is a crucial task before entering other analyses. Cleaning may remove inconsistent and misleading observations so an efficient data cleaning may be a factor for a successful project. Our case here does not present obvious cleaning chances. Nevertheless, an investigation in the store id variable showed that the unique stores in the training set are 1115 while the unique stores in the test set 856. This means that the training set contains information for 259 stores not presented in the test set. The information of extra 259 stores in the training data not only is redundant but also may deceive the model. So, in this stage, we found all of these observations and we removed it from the training set which led to a training set of 773,231 observations instead of 1,017,209. So, an extra gain from cleaning the data is that the next processes will need fewer computational power since the amount of data decreased by more than 20%.

Treat Missing Values.

For a dataset to present missing values is something that may occur often and it is mostly the rule rather than the exception. For a data scientist treating missing values is a challenging

task and if completed successfully will allow the model to exploit the dataset's full potential. This dataset presented missing values both in the store set and in the test set. The train set does not contain any missing values and this is a pleasant surprise.

Treat Test Set.

The test set present 11 missing values in the Open variable (this indicates whether a store is open or not). Also, it misses all values from Customers number which was expected anyway, because it is very logical to not know how many customers will come into the store in the future. The number of customers and sales are so closely correlated that knowing the one is almost the same as knowing the other. That also means that we cannot use customers in our predictive model except if using lagged observations. Nevertheless, our choice for this project is to not use any lags either customers or sales. So we focus on the 11 missing values in the Open variable.

We printed these eleven observations and we found that none of them is a state holiday, a school holiday, or a Sunday. Moreover, all of these observations appear on one unique store id 622. We investigated also how this store behaves in the train set and found no indication that the store can be closed on other days than the indicated holidays or Sundays. So we assumed that the store 622 will be open on these 11 days and replaced the missing value with 1.

Treat Store Set.

The stores' set misses 3 observations describing the competition distance of the stores and 354 observations lack the information about how much time the competition exists. We followed a usual approach here and we replaced all these missing values with the medians of the distributions which were: 2325 for competition distance, 2010 for competition year and 8 for competition month.

The Promo2 variables also miss 544 observations from its three relative features (PromoYear, PromoWeek, PromoInterval) on the exact same dataframe index. Here we followed a different approach. The Promo2 variables include also a variable of interval (PromoInterval) in which the promotions run. What seems logical here is a promotion not existed at all in the missing values and that is why there is not a value there. So since the missing value indicates a non-promotion for this store we should find a replacement for the year and week of the observation indicating non-promotion. That is equal to zero the promotion effect for this variable. To achieve this zero effect we replaced the promotion dates to be exactly after the end of the examined dates in the test set. These values are 2015 for the year and 39 for the week since our last data date is 2015-09-17 which corresponds to the 38th week of the year.

Transform – Generate – Extract Features

Transforming, extracting and generating new features from the already existed ones is a data science art related to creativity, domain knowledge and statistical literacy. The purpose of this

task is to utilize features like text or dates and increase the predictability of numerical or categorical features by applying transformations or finding interactions. The data of this project gives the chance to use the above methods as it presents date and text features needing extraction, categorical data that could generate interactions and then some numerical features that could be transformed. The nature of the examined data permits at this point to merge all the datasets and perform the process once and then we can split again the data in the train and test sets.

Store Type - Assortment.

Store Type and Assortment are two categorical features describing the nature of the stores. Both of these features present three values. We took the interaction of the two categories to assemble a new feature which we named StoreTypeAssortment. This means that the category 'a' of the StoreType and the category 'b' of the Assortment created the category 'a_b' for the StoreTypeAssortment.

Promo2SinceYear – Promo2SinceWeek.

The feature Promo2SinceYear defines the year on which the Promo2 started and the Promo2SinceWeek defines the week the Promo2 started. These features are redundant and we combined them in one. Our concept of the combination is to make one feature describing the total number of weeks the promotion started. To achieve that we deducted the year of each observation from the maximum year of the observations and then we multiplied the outcome by 52, transforming that way the years since the promotions in weeks. Then we added the weeks from the Promo2SinceWeek variable and removed the two old features.

CompetitionOpenSinceYear- CompetitionOpenSinceMonth.

The features of CompetitionOpenSinceYear and CompetitionOpenSinceMonth describe when the competition was born. These features are redundant and we treated them like the Promo2 similar features. We multiplied the outcome of the deduction of the year of each observation of CompetitionOpenSinceYear from the maximum year with 12 and we added the months from the CompetitionOpenSinceMonth and then we removed the old features.

PromoInterval.

The feature PromoInterval is now a text feature containing months abbreviations separated by a comma like "Jan, Feb, Mar". This kind of representation is not of any use for the model since textual features must be transformed to numerical before they enter a Machine Learning algorithm. Our idea about this feature is to represent the values in a way that they could be compared to the month of the observations and if the month of the observation is in the months of the interval then set this PromoInterval value as 1 else set it as 0. To achieve this

binarization we created a python dictionary that maps the months' abbreviations to months numbers and we transformed the textual feature to an array containing the months' numbers. Then we compared this list to the extracted month from the date feature and binarized the result of the extracted month being in the array.

Extracting Datetime Features and Adding Polynomials.

The data as it is now unified in one dataset contains a DateTime feature from which we easily extracted 4 new features using pandas methods. These are the Year, the Month, the Week and the Day. Having already one more feature the DayOfWeek, the total number of the DateTime features are now five. Then we normalized the Year feature by deducting the minimum year of the distribution. This also proves helpful if wanting to add polynomials using the year.

To illustrate the seasonality of the data we decided to add polynomials of the DateTime features until the 4th grade. This way, we added 12 new features to the data (0 grade is redundant and 1st grade already exists so we added 2nd, 3rd, and 4th grades). Adding polynomials in a regression model may lead to overfitting and multicollinearity, so a feature selection here will be useful to keep only the necessary features that really add information to the model.

One-Hot-Encoding.

Our data set contains now several categorical features that cannot be used by a regression model. One way to transform the categories into numbers is by one-hot-encoding. This method creates one new feature for every unique value of every categorical feature and then binarizes the presence of every value in every observation. The drawback of this method is that adds extensive dimensionality to the data especially when the categorical features are steeped in categories.

Unfortunately, our data contains 856 stores which means that if we encode them using this method we would add $856 * 776,531$ values to the dataset. A different way could be to encode the stores using Label Encoding but this way the algorithm will treat the encoding like numbers interacting one with each other, something that could either mislead the model or at the best case zero the effect of the feature on the regression. We also considered that features like competition distance and store-assortment interaction may provide enough separation of the observations among different stores. Thus, we decided to keep the stores' one-hot-encoding for the final predictive model but run the analyses and the feature selection without it.

The final list of features on which we applied one-hot-encoding is StoreType, Assortment, StoreTypeAssortment, SchoolHoliday, StateHoliday and Store IDs. With this transformation, the preprocessing ends with a full dimensionality of 889 features including also the one-hot-encoded-stores.

Exploratory Data Analysis (EDA)¹

EDA is a data science task in which the scientist explores the data and reveal insights. Though tables can be used in this task, the main strategy of EDA is to visualize the variables trying to spot patterns inconceivable when looking at data or reading tables. The experience and the domain knowledge of the scientist can prove decisive for this task and the result might help to build a better predictive model, generate more features, or apply an unusual transformation to the data. Moreover, EDA can be a part of data analysis and is useful for managers to obtain an idea of what their data might reveal to them. Our dataset consists of several features that present interest in their relation to the target variable as well as their relation to one another. Specifically, we focused on exploring the connection of sales to competition, promotions and time.

Sales in Time.

We summarized the sales of the stores daily and monthly and visualized how sales are distributed in two separate line plots (Figure 2). The daily plot shows that the sales deviate over time and the signal presents several peaks and troughs. The monthly sales show obviously that the level of sales dropped significantly in 2014 presenting a trough in October of that year but sales start raising again in 2015. The peak of the sales was in 2013.

State Holiday.

The days of Easter and Christmas are state holidays (Figure 3) and stores have almost zero sales. Figure 3 shows that on these holidays sales are zero or near zero.

Impact of Promo & Promo2.

Promotions are very important to sales. The time and kind of promotion may have an impact on sales for a long time. In this data (Figure 4), seems that promotions correlate to sales since 2014 which is a bad year for the sales Promo and Promo2 runs in fewer stores than previous and next year. Focusing on Figure 4 exactly after June of 2014, we see that running Promo and Promo2 in fewer stores falls exactly in the sales trough.

Another approach of analyzing Promo is indicating the promotions sums in the relation of two other variables (Figure 6). The notion one can get from Figure 6 is that more promotions are positively correlated to Customers and Sales.

Competition and Sales.

The usual idea about competition is that competition harms sales. A logical idea in the head of managers is to avoid competition being at a close distance and that old firms could have

a negative effect on the sales of the newer ones. We analysed two competitions levels, the competition distance from a store and the competition age (Figure 5). We used each variable as an explanatory variable of a univariate regression where the target is the sales and found that the impact of the explanatory variable on the regression is very low. We can say the notion of the harming competition is not confirmed in this data.

Correlation Analysis¹

Correlation analysis is used to estimate the linearity strength of two numerical variables. We expect two highly correlated variables to produce prediction close to reality when using one of them as an explanatory variable and the other one as a response in univariate regression. We use the Person correlation to estimate how much each explanatory variable connects to sales but also to reveal explanatory variables that are correlated one to another and could cause a multicollinearity effect to our regression. We summarize our results in a heatmap (Figure 7) and calculate the most correlated variables (Table 2). From Figure 7 and Table 2 we can see that the sales are mostly positively correlated to the Open variable (open store equals more sales), and mainly negatively correlated to the day of the week (closer to Sunday equals fewer sales). A possible multicollinearity issue may be created by the DateTime features which are highly correlated to each other. The only correlation to be characterized as a strong one is the Open variable while the correlations of the day of the week and the Promo to sales are medium correlations.

Feature Selection¹

Regression Analysis.

Regression Analysis is a method of defining which variables are important to keep in a model and which are not. Moreover, the coefficients (slopes) of the regression equation define the impact of a variable. The statistical significance of one variable can be determined by the p-value of a t-test where the null hypothesis (H0) is that the coefficient of the variable in the regression equation is zero. We determine a p-value threshold under which the H0 must be rejected. Usually, scientists use the 0.05 p-value threshold and we follow this line in this study. We used the statsmodels OLS wrapper of Multivariate Linear Regression (MLR), which is implemented also in Python.

Lasso Regression.

Lasso Regression is a linear regression using regularization. Regularization in Lasso shrinks coefficients of the regression until zeroing them using a parameter α to define regularization intensity. The kind of regularization is L1 meaning that it uses the mean squared error to define the loss of a model. From the scikit-learn lasso regressors, we selected a variant of Larsson which also perform cross-validation to calculate the loss of the model and to select the best α value (LLR). Lasso regression is an alternative method of performing feature selection since after the model is trained, several variables may have ended with zero coefficients. Lasso Regression needs to have the explanatory variables normalized because it penalizes the large coefficients. The Lasso wrapper we use contains also an automated normalizing parameter.

Feature Selection Pipeline.

We combined both of these methods and used our data without having stores encoded and without polynomials of DateTime features. We trained three MLR models and one LLR. The pipeline of the analyses is to test the MLR without any feature selection (MLR1), then add the polynomial features and train an LLR with 10-fold cross-validation for selecting the best α value among 1000 values, then perform the feature selection of LLR and train again an MLR (MLR2). Next step is to remove the features still having a p-value greater than 0.05 and then train again an MLR (MLR3) and confirm that MLR3 provides the best structure of features since it presents the highest adjusted R^2 and the lowest AIC-BIC also using the fewest features.

Model Selection¹

After having performed the feature selection, we need to explore the regressors that could be our model to predict the test set. We decided to compare two algorithms: a Multivariate Linear Regression (MR) and a Ridge Regressor (RR). For the selection of the best model, we create a validation dataset that consists of the last month observations of the training set. We use the regressors provided from the *linear_model* of the python's *sklearn* library which also contains a Ridge Regressor wrapping k-fold cross-validation. After selecting the most appropriate model which proved to be Ridge Regression we predicted the test data.

Ridge Regression.

Ridge Regression like Lasso Regression penalizes coefficients of the regression using an α parameter to define regularization. The regularization of Ridge Regression is given by the L2 norm so minimizes the objective function $\|y - Xw\|^2 + \alpha * \|w\|^2$. We selected the following α values to be tested by a 10-fold cross-validation: $[1e-7, 1e-6, \dots, 1e1]$. The wrapper also provides a parameter to auto-normalize the explanatory variables since Ridge like Lasso penalizes the large coefficients so not normalizing might skew the provided weights of the equation.

Evaluation.

To evaluate MR and RR we use the Correlation Coefficient (R^2), the Mean Absolute Error (MAE), the Mean Absolute Percentage Error (MAPE), the Root Mean Squared Error (RMSE) and the Root Mean Squared Percentage Error (RMSPE). Metrics like R^2 and MAE are implemented in sklearn but we implemented the functions of RMSPE, MAPE and RMSE.

Results

Data provided us with 6 explanatory variables (without customers and stores) which after performing feature extraction and feature generation we managed to enter the feature selection pipeline with 48 features from which we kept the 35. During the feature selection process we managed to increase the adjusted R^2 from 0.59 which was the result of MLR1 to 0.62 which was the result of MLR3 using 13 features less.

We used a validation set consisted of July 2015 (856 rows) to compare the predictability of MR and RR. The dimensionality of the models was 892. The results did not leave any doubt about the importance of regularization. The MR model had almost twice as better results on all metrics (Table 3) but is enough to mention the R^2 metric which provided a 0.74 score for RR in comparison to a negative -0.01 for MR. So while RR accounts for the 74% of the variability of the sales the MR is not better than the mean of sales.

The graphical analysis of the predictions and their residuals (Figure 9) on the validation set showed that the model overestimates the sales when they are close to 0 but underestimates them when they are more than 15,000. The residual plot shows clearly that residuals have a trend. The mean of the residuals is calculated at -584 and their trend can be given by the equation $\varepsilon = 0.298\hat{y} - 3043$.

Analyzing the size of the regression coefficients (Figure 10) we see that the store type b with the assortment a provides the most positive relation to the sales with a coefficient close to 8,000, followed by the Open binary variable with a coefficient close to 7,000. The only negative variable worth mentioning is the day of the week which present a negative coefficient close to -7,000 but this impact is being balanced by the 2nd degree polynomial of this variable which presents a positive coefficient of about 2,000.

The distribution of the predictions in the test set presents similarities and differences with the sales in the train set (Table 4). The mean and the standard deviation can be considered similar but the predictions present several negative values that needed to be clipped to the minimum of zero. The maximum predicted sales are almost half of the maximum ones of the train set but this could be due to the examined period of the test set which lacks months with high sales. The clipped predictions present a sales signal that can be considered normal and expected (Figure 11).

Conclusion

In this study, we preprocessed and analyzed a demanding dataset of about one million rows and constructed a predictive model which utilizes the high dimensionality we produced during the preprocessing. We performed a feature selection combining Lasso Regression and Regression Analysis and dropped 13 of the variables before entering the construction of the predictive model. We trained two Multivariate Linear Regressors, one without any regularization and one with ridge regularization optimizing the alpha parameter with 10-fold cross-validation. We found that only the ridge model is efficient for this kind of data validating the ridge result with a validation set. The RR model achieved a decent 0.74 score of the correlation coefficient and a MAPE of about 12% in comparison with the MR model which achieved statistically insignificant results. Moreover, we dealt with a test set to which we predicted the expected sales and confronted the challenge of dealing with a bounded variable like sales.

The accuracy of the predictions in the test set is dependent on the predictors to appear a similar behaviour in the test set and train set. While the prediction process in the validation set did not produce any negative values, the prediction process in the test set, presented several negative values which can be interpreted as a violent change in the predictors' distributions. So it will not be odd to achieve a lower score in the test set than the one in the validation set.

Several improvements may occur, yet from the validation set. We saw that the residuals in the validation predictions present a trend and the mean of the residuals is relatively distant from zero. We could further try to improve the model by transforming the sales or/and other variables to be closer to the normal distribution. We could also try to generate more features and tailor behaviours of the examined periods in the model or the results. We could also directly remove the trend from the predictions which would immediately improve the result in the validation set. Nevertheless, even from the validation set, we can expect issues on the zero sales as the model fails to predict efficiently the sales when they are very low or zero. This kind of problems seems common when dealing with bounded variables which are prone to present high concentrations of values near the bounds.

We conclude this study with the conclusion that regularization and feature selection can be the secret of a successful model and data cleaning and preprocessing are fundamental to start from significant basis.

Works Cited

- [1] Pedregosa *et al.*, (2011), *Scikit-learn: Machine Learning in Python*, JMLR 12, pp. 2825-2830, 2011.

Tables

Table 1

a) Basic Statistics of Training Set

	Store	DayOfWeek	Sales	Customers	Open	Promo	SchoolHoliday
count	1017209	1017209	1017209	1017209	1017209	1017209	1017209
mean	558.4297268	3.998340557	5773.818972	633.1459464	0.830106694	0.381514517	0.178646669
std	321.9086511	1.997390965	3849.926175	464.4117339	0.375539225	0.485758605	0.383056368
min	1	1	0	0	0	0	0
25%	280	2	3727	405	1	0	0
50%	558	4	5744	609	1	0	0
75%	838	6	7856	837	1	1	0
max	1115	7	41551	7388	1	1	1

b) Basic Statistics of Test Set

	Store	DayOfWeek	Open	Promo	SchoolHoliday
count	41088	41088	41077	41088	41088
mean	555.8995	3.979166667	0.85432237	0.395833333	0.44348715
std	320.2745	2.015481292	0.352787029	0.489034893	0.496802078
min	1	1	0	0	0
25%	279.75	2	1	0	0
50%	553.5	4	1	0	0
75%	832.25	6	1	1	1
max	1115	7	1	1	1

c) Basic Statistics of Stores Set

	Store	CompetitionDistance	Promo2
count	1115.00	1112.00	1115.00
mean	558.00	5404.90	0.51
std	322.02	7663.17	0.50
min	1.00	20.00	0.00
25%	279.50	717.50	0.00
50%	558.00	2325.00	1.00
75%	836.50	6882.50	1.00
max	1115.00	75860.00	1.00

Note: Basic Statistics of the three datasets.

Table 2

Positively Correlated	Correlation	Negatively Correlated	Correlation
<i>Open</i>	0.69	<i>DayOfWeek</i>	-0.45
<i>Promo</i>	0.45	<i>StateHoliday_a</i>	-0.21
<i>StateHoliday_0</i>	0.26	<i>StateHoliday_b</i>	-0.12
<i>StoreType_b</i>	0.17	<i>StateHoliday_c</i>	-0.09
<i>StoreTypeAssortment_b_a</i>	0.15	<i>Assortment_a</i>	-0.08
<i>StoreTypeAssortment_b_c</i>	0.12	<i>Promo2</i>	-0.08
<i>Assortment_b</i>	0.08	<i>StoreTypeAssortment_a_a</i>	-0.07
<i>StoreTypeAssortment_b_b</i>	0.08	<i>Promo2Weeks</i>	-0.05
<i>StoreTypeAssortment_a_c</i>	0.08	<i>StoreTypeAssortment_d_a</i>	-0.04
<i>SchoolHoliday</i>	0.08	<i>IsPromo2</i>	-0.04

Note: This table presents the 10 most positive and the 10 most negative correlation of predictors to the target.

Table 3

	MR	RR
mae	2149.39	1101.78
mape	31.94	16.80
rmse	2762.28	1393.02
rmspe	43.11	23.96
r2	-0.01	0.74

Note: The table presents the results of a Multivariate Linear Regression (MR) and the Ridge Regression (RR) on the validation test of the data.

Table 4

	Train	Test prediction
count	772375	41088
mean	5688.69768	5691.206482
std	3723.68901	3307.587636
min	0	-4997.445321
25%	3776	4291.769715
50%	5697	6030.384359
75%	7722	7641.373305
max	41551	23563.40608

Note: The table is a comparison of the basic statistics of the sales in the training set and the predictions of the model in the test set.

Figures

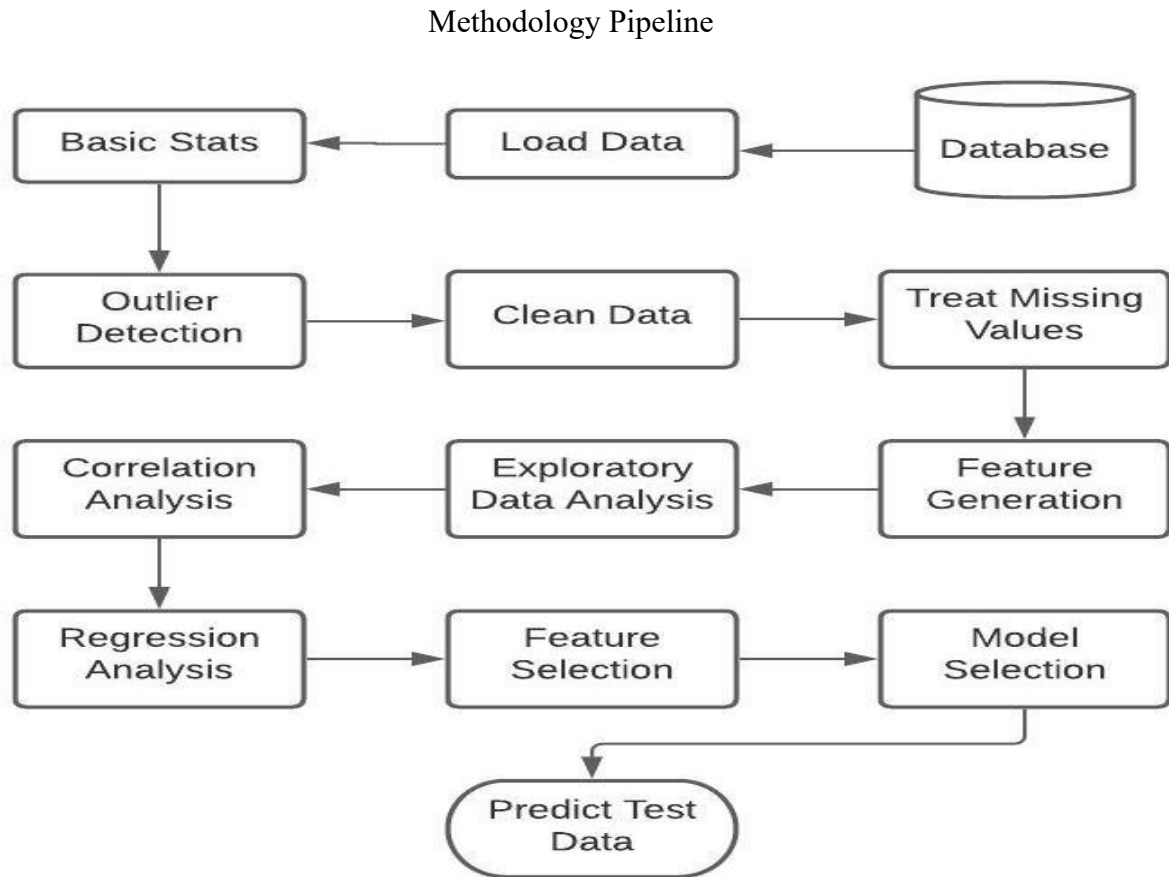


Figure 1. The figure indicates the order of decisions and actions taken in this project on the highest level.



Figure 2. The year 2014 was a bad year for sales. Daily data presents a signal that deviates intensively while monthly seasonality is skewed by the sales trough in 2014.

Sales on Holidays

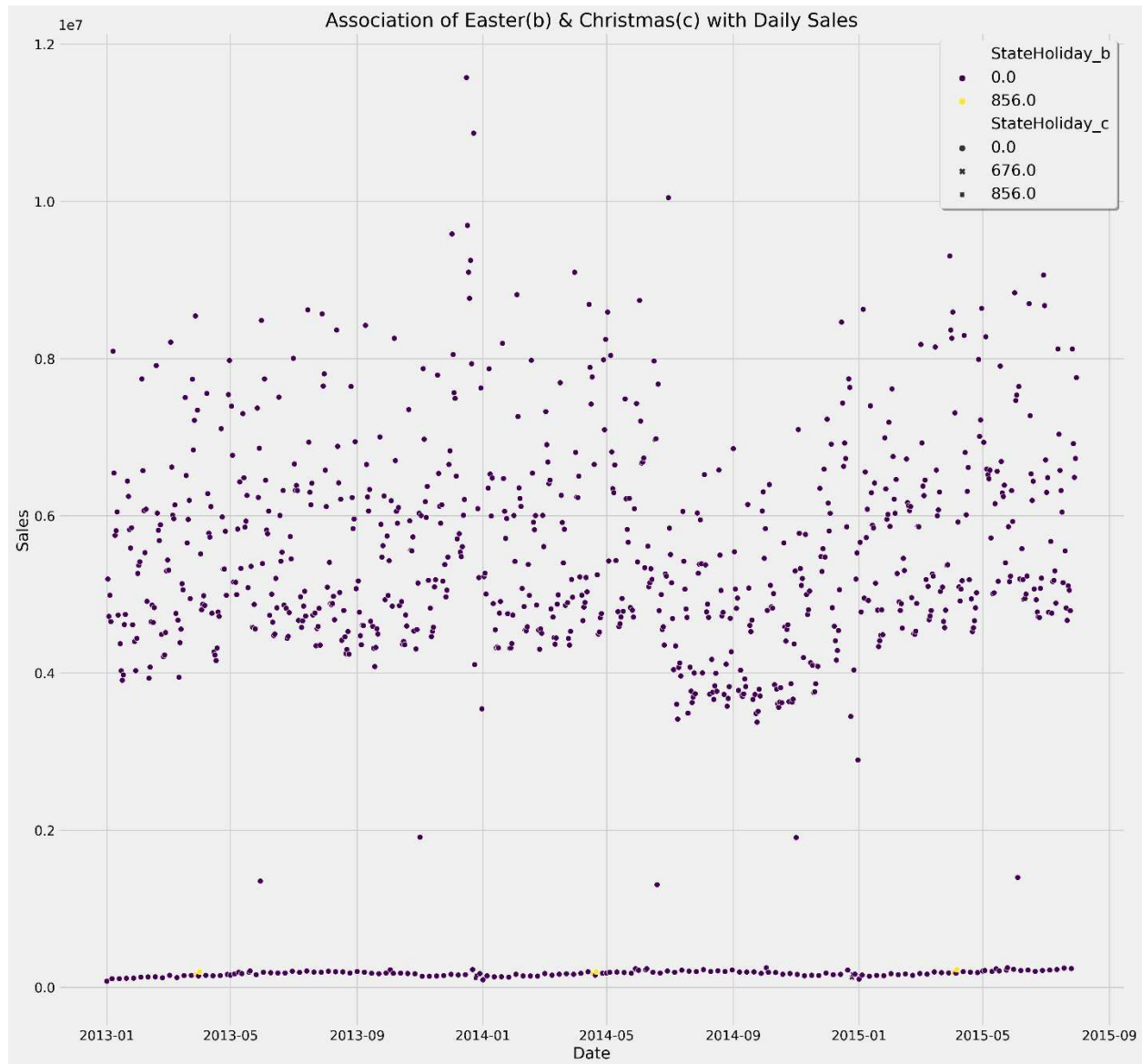


Figure 3. The scatterplot indicates the height of daily sales in the data for different stores. When a marker is yellow means that all 856 stores have the Easter Holiday. When the marker is X or a square means that 676, or 856 stores have Christmas holiday.

Impact of Promotion on Sales

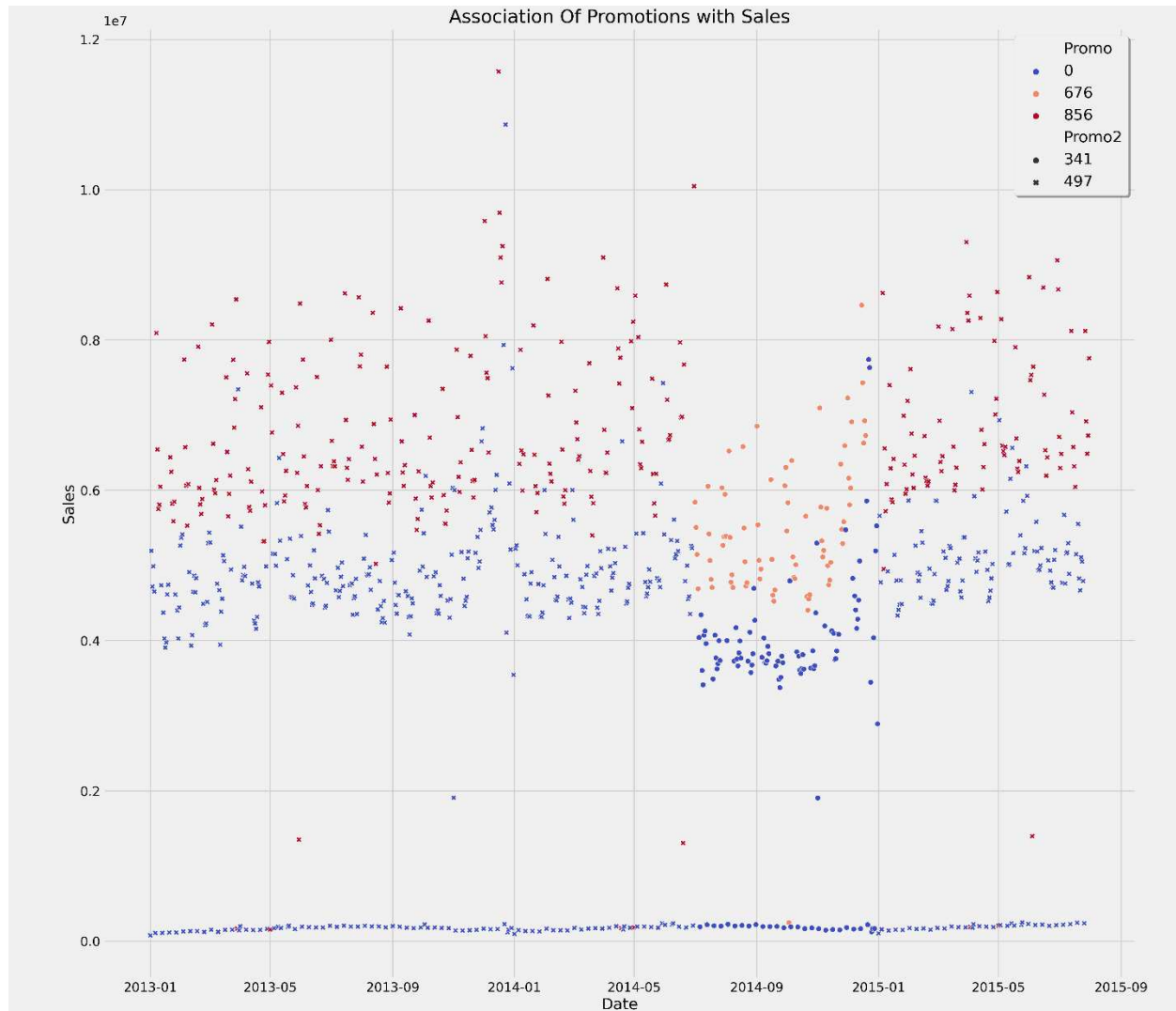


Figure 4. The scatterplot indicates the height of daily sales in the data for different stores. When a marker is blue means that no stores have a Promo running. If the marker is orange or red then 676 or 856 stores have the Promo running, respectively. When the marker is circle 341 stores have Promo2 running while when the marker is an X, 497 stores have Promo2 running. This graph can display combinations: When a marker is a red X then 856 stores (all in the train set) run Promo and simultaneously 497 of them run also Promo2.

Competition Impact on Sales

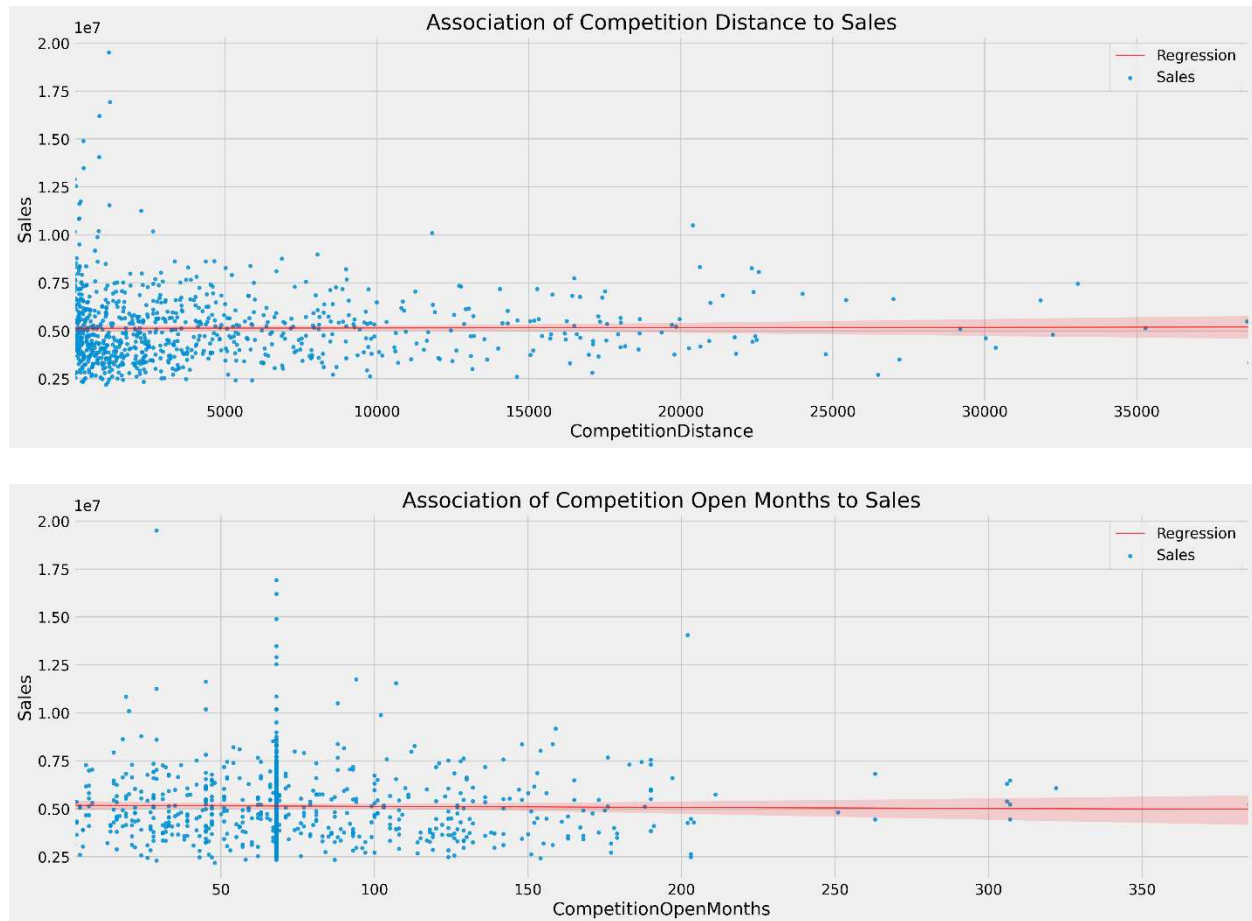


Figure 5. The two competition graphs present the univariate regressions of Sales against competition distance (upper graph) and the number of months of competition existence (bottom graph).

Promo in Pairplot

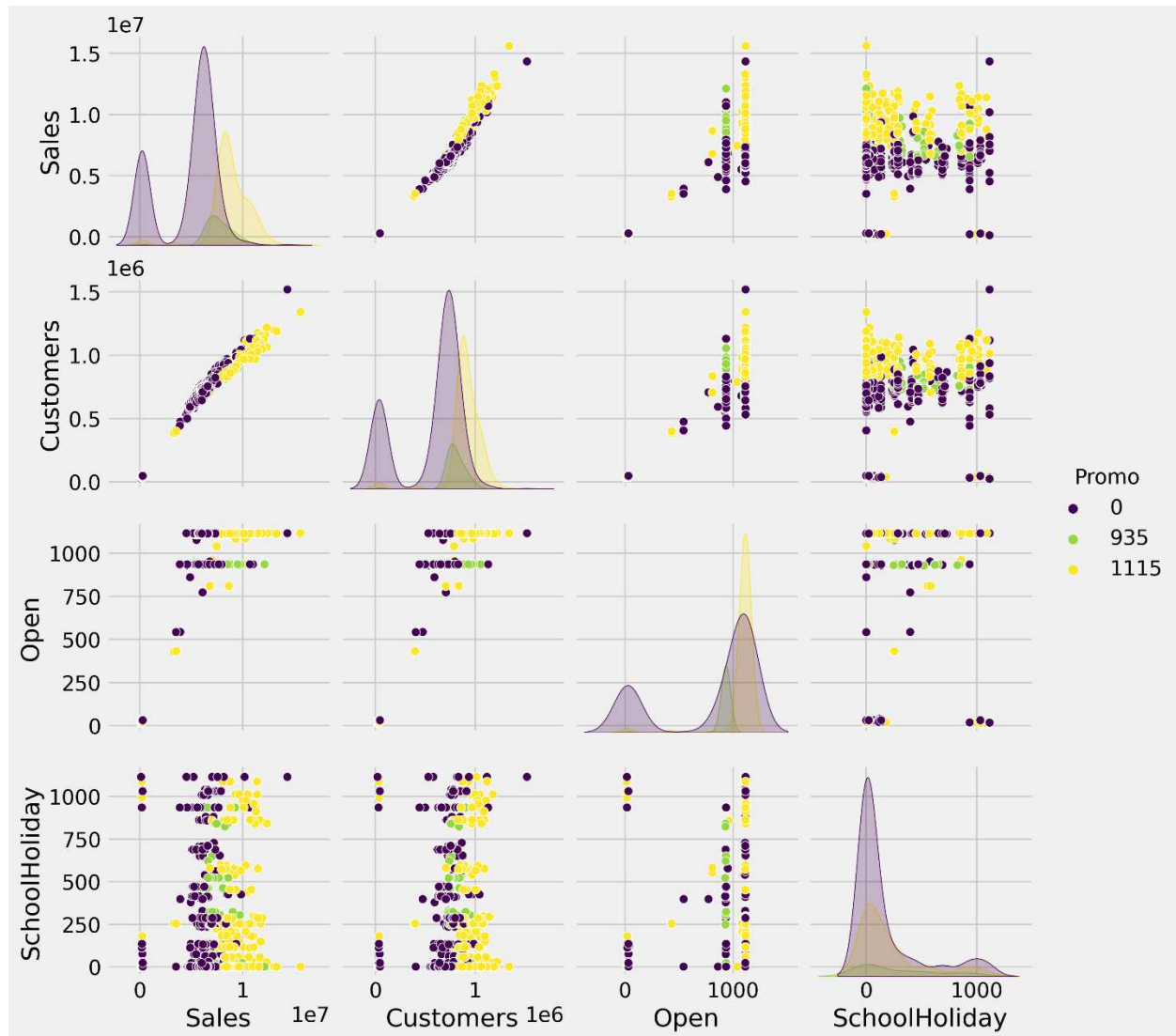


Figure 6. This comparative plot analyzes how Promo is distributed in the analysis of two other variables. This plot takes account of the full train set before cleaning the data from store ids and customers. The colours in the plots indicate the sum of the stores which run a Promo daily.

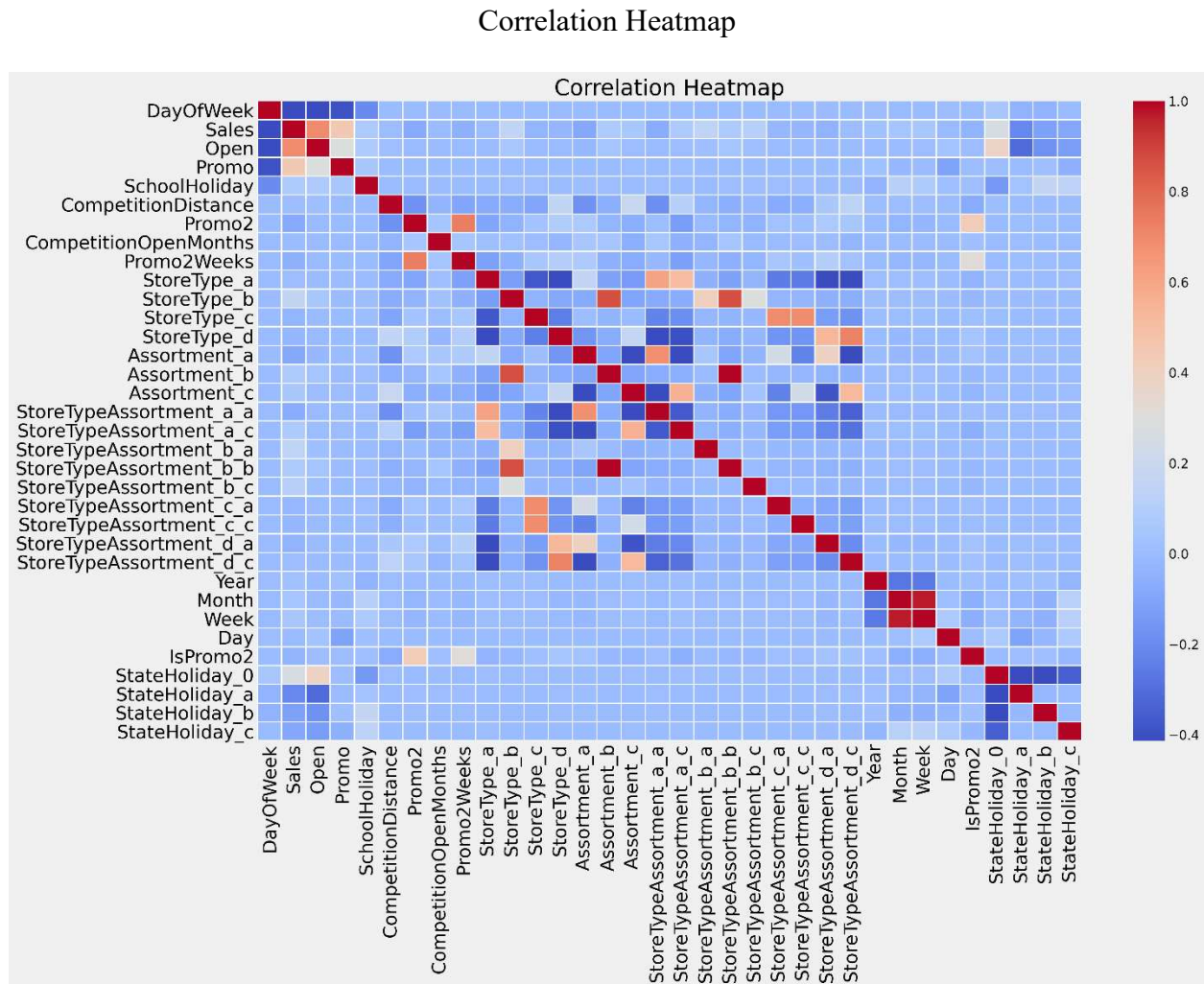


Figure 7. This heatmap describes the correlation sign and correlation intensity between two variables. The colouring can be interpreted as: the deeper the blue the highest the negative correlation, the redder the highest the positive correlation and the closer to white (or the less intensive the colour is) the weaker the correlation.

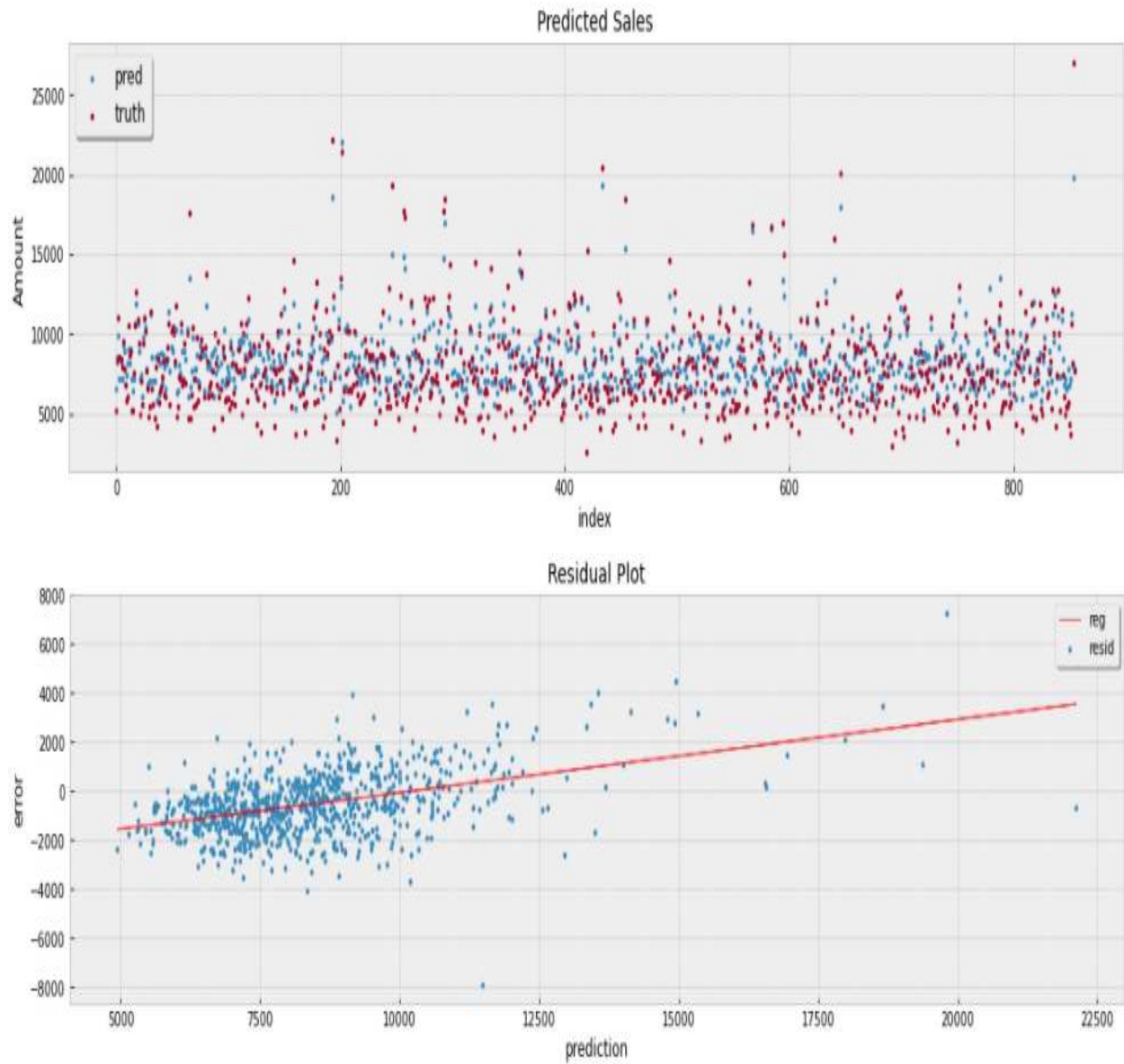


Figure 8. The upper plot shows the deviation of the predictions of the validation set from the true values and the bottom plot shows the distribution of the residuals and their trend.

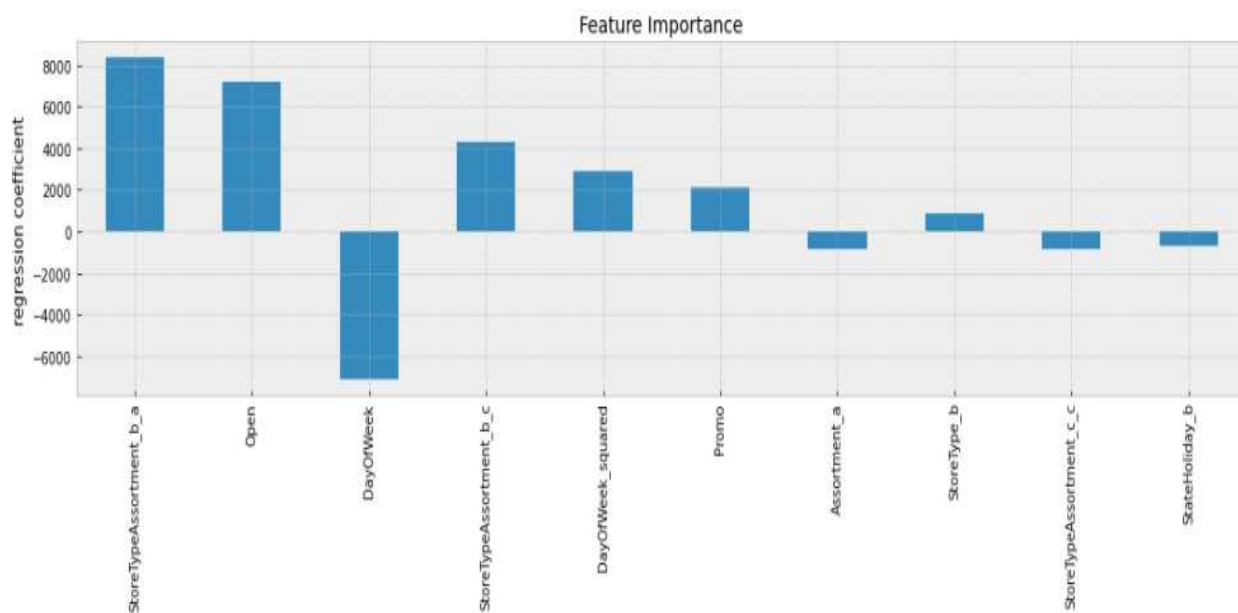


Figure 10. The plot shows the larger regression coefficients and their sign.

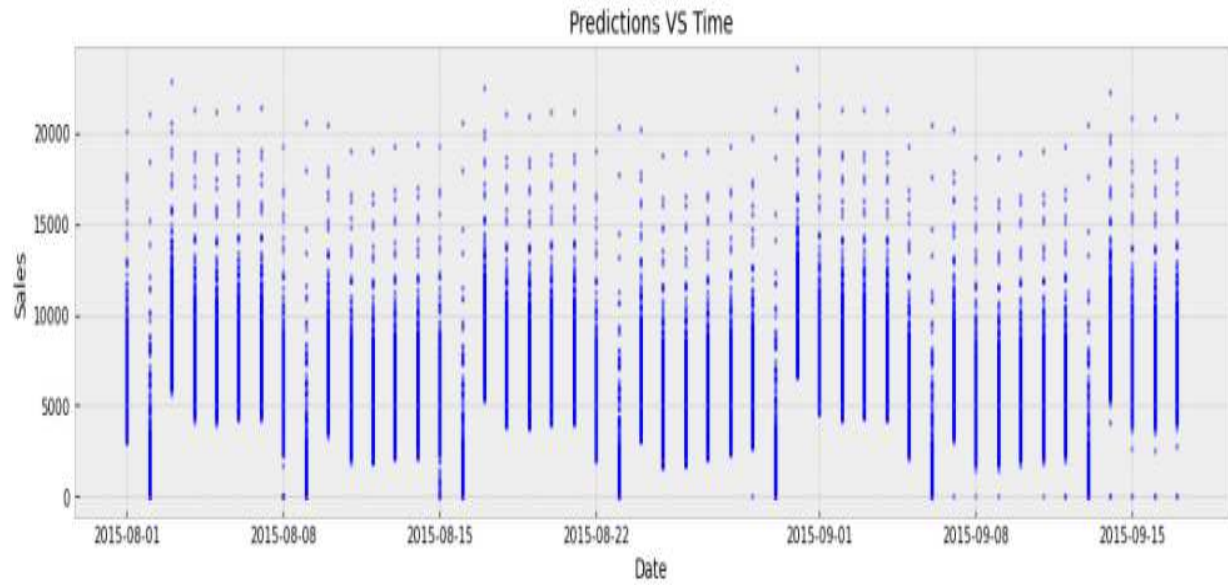


Figure 11. The plot shows the predictions in the test set after adjusting the negative sales.